Curvature and Torsion in the Continuum Theory of Lattice Defects

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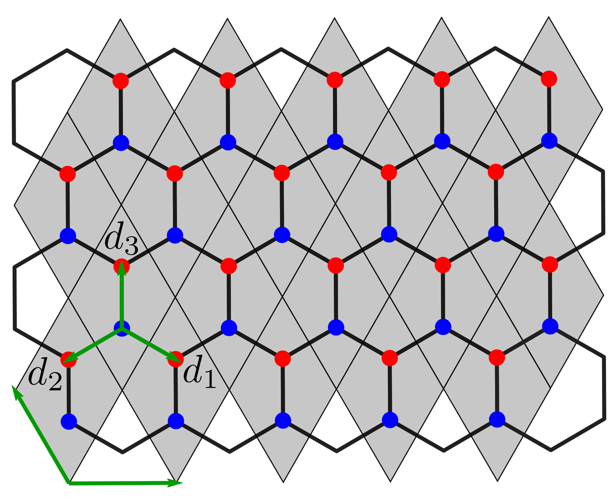
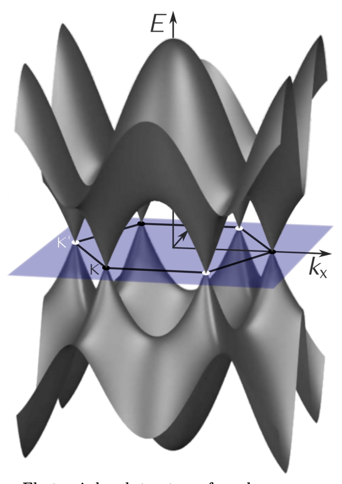
Motivation

- Current flow paths for nanoelectric devices with simpler model than tight-binding
- Defect arrangements might allow for new types of nanoelectric logic
- Better understanding of fundamental representation of defects and dislocations in the continuum theory of graphene.

Continuum Theory - Graphene[1]





Electronic band structure of graphene.svg. (2025, November 29). Wikimedia Commons. Retrieved February 18, 2026

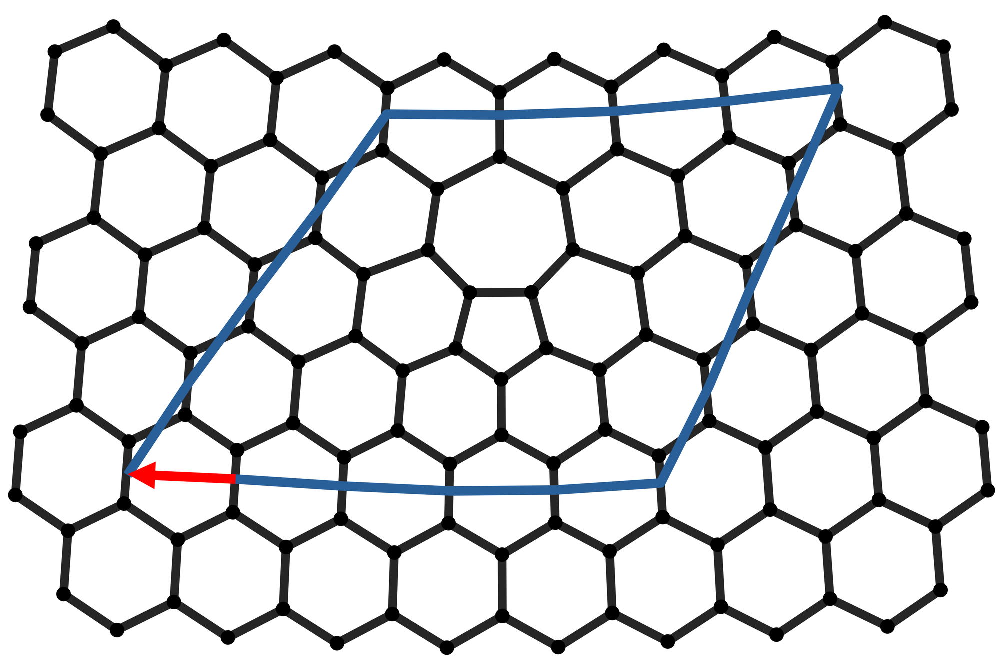
Tight-binding Hamiltonian

$$H = \sum_{\vec{r}} \sum_{j=1}^{3} \left(a^\dagger_{\vec{r}} b_{\vec{r}+\vec{d}_j} + b^\dagger_{\vec{r}+\vec{d}_j} a_{\vec{r}} \right)$$

Continuum limit

$$i\partial_t \vec{\phi}(\vec{r}) = i\, c\, \boldsymbol{\sigma}^i \partial_i \vec{\phi}(\vec{r}), \qquad \rho = \vec{\phi}^\dagger \vec{\phi}, \qquad \vec{j} = -c\vec{\phi}^\dagger \vec{\boldsymbol{\sigma}} \vec{\phi}$$
$$\Updownarrow$$
$$i\gamma^\mu \partial_\mu \phi = 0, \quad \gamma^\mu = \sigma^\mu \sigma^3, \quad \mu \in \{0,1,2\}$$

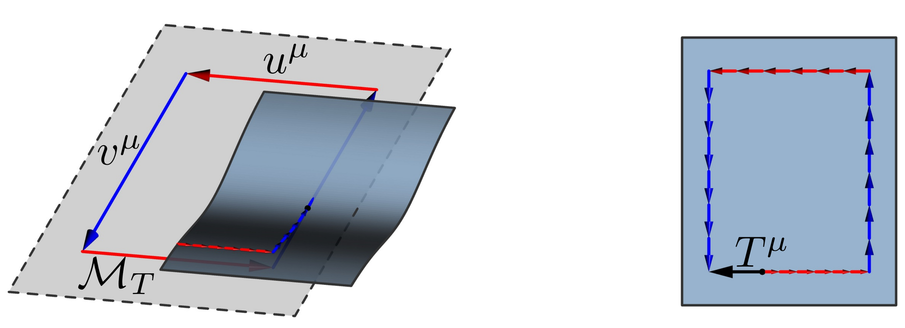
Burgers vector for a 5-7 defect[2]



- **Lattice dislocation** with Burgers vector:
$$\vec{b} = -\sqrt{3}\, \vec{u}_x$$

Torsion[3]

- Standard GR: torsion vanishes
- Torsion = **antisymmetric** part of the connection
$$T^\lambda{}_{\mu\nu} = \Gamma^\lambda{}_{\mu\nu} - \Gamma^\lambda{}_{\nu\mu}$$
- Decomposition of any general affine connection into symmetric part, contorsion & disformation:
$$\Gamma^\alpha{}_{\mu\nu} = \{{}^\alpha{}_{\mu\nu}\} + K^\alpha{}_{\mu\nu} + L^\alpha{}_{\mu\nu}$$
- **Metric-compatibility**:
$$\nabla_\lambda g_{\mu\nu} = 0 \Leftrightarrow L^\alpha{}_{\mu\nu} = 0$$
- **Contorsion tensor** from torsion:
$$K^\alpha{}_{\mu\nu} = \frac{1}{2}\left(T^\alpha{}_{\mu\nu} - T_{\mu\nu}{}^\alpha + T_\nu{}^\alpha{}_\mu\right)$$
- **Visual** idea: displacement vector $T^\mu(u,v)$ when rolling tangent plane along parallelogram:
$$T^\mu(u,v) = u^\lambda v^\nu T^\mu{}_{\nu\lambda}$$

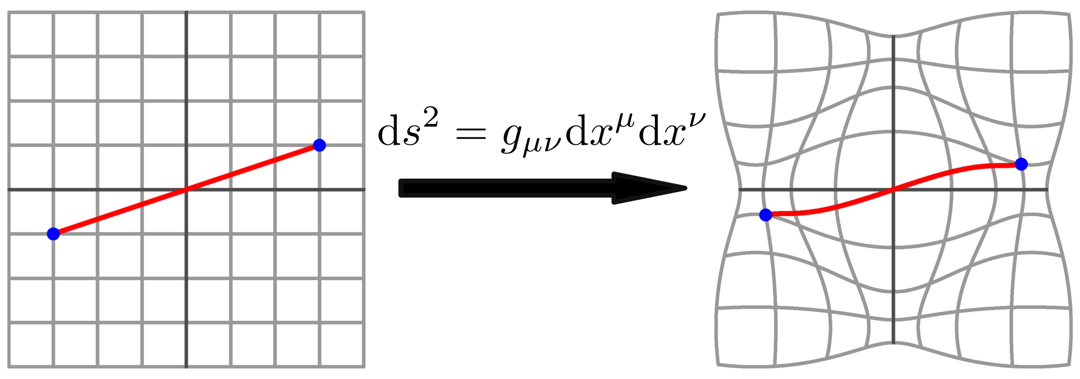


Riemannian Geometry[3]

- Riemannian **manifold** $\mathcal{M}$ with metric $g_{\mu\nu}$ and connection $\Gamma^\lambda{}_{\mu\nu}$

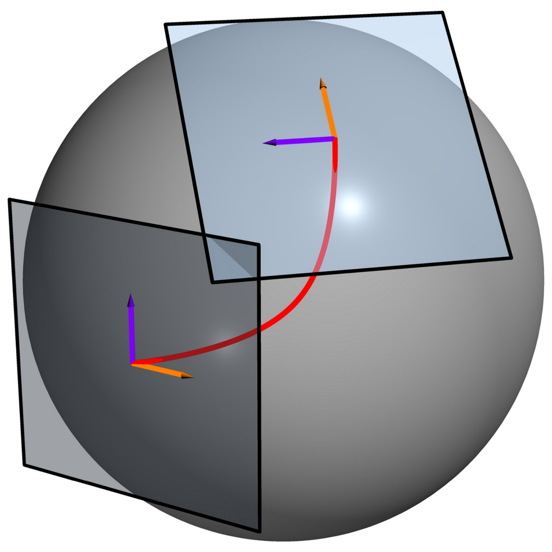
Metric

- Defines distances & angles on manifold



Connection

- Connectes nearby **tangent spaces**
- Allows definition of **tensorial** derivative
- Unique torsion-free, metric-compatible connection: **Levi-Civita connection**



$$\{{}^\lambda_{\mu\nu}\} = \frac{1}{2} g^{\lambda\rho} \left(\partial_\nu g_{\rho\mu} + \partial_\mu g_{\rho\nu} - \partial_\rho g_{\mu\nu}\right)$$

Covariant derivative

- Takes into account the **structure** of the manifold
- Transforms like a **tensor**
$$\nabla_\mu v^\nu = \partial_\mu v^\nu + \Gamma^\nu{}_{\mu\lambda} v^\lambda$$

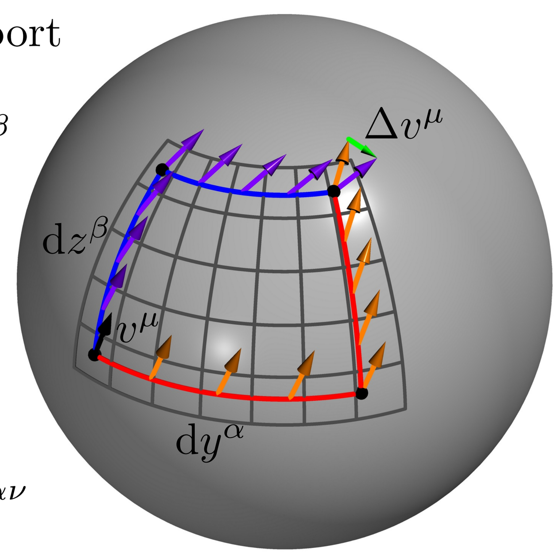
Autoparallels

- Tangent vector is parallel-transported
$$u^\mu \nabla_\mu u^\alpha = 0$$

Curvature

- how does parallel transport depend on the path?
- $\Delta v^\mu = R_{\alpha\beta}{}^\mu{}_\nu\, v^\nu\, dy^\alpha\, dz^\beta$
- Fully determined by the connection:

$$R_{\alpha\beta}{}^\mu{}_\nu = + \partial_\alpha \Gamma^\mu{}_{\beta\nu} - \partial_\beta \Gamma^\mu{}_{\alpha\nu} + \Gamma^\mu{}_{\alpha\lambda}\Gamma^\lambda{}_{\beta\nu} - \Gamma^\mu{}_{\beta\lambda}\Gamma^\lambda{}_{\alpha\nu}$$



Metric ↔ torsion mapping in 2D

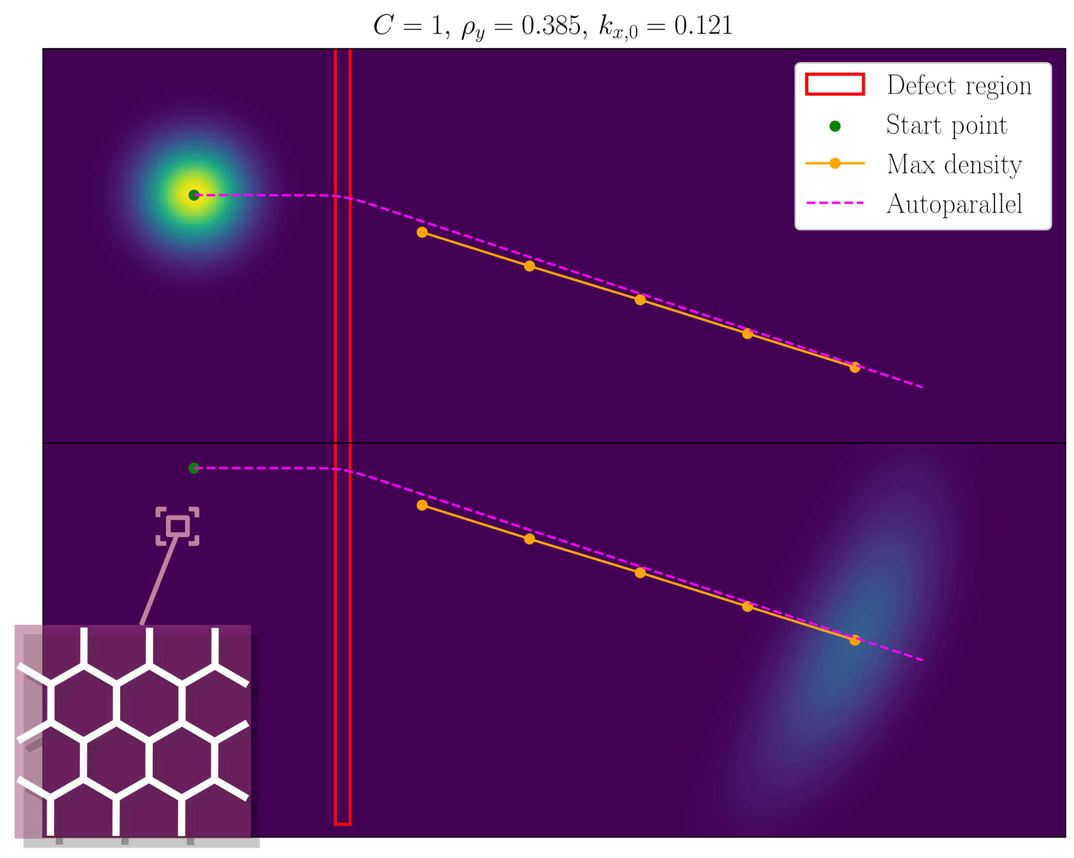
- Physical mapping between:
  - **Classical** Manifold $\mathcal{M}_g(g_{ij}, \{{}^k_{ij}\})$
  - **Purely torsionful** Manifold $\mathcal{M}_T(\delta_{ij}, K^k{}_{ij}\})$
- Isothermal coordinates: $g_{\mu\nu} = \delta_{\mu\nu}\, e^{2\phi(x,y)}$.
- Antisymmetry: $T^k{}_{ij} = b^k \varepsilon_{ij}$
- Demand equal **curvature** of both manifolds:
$$R\left[\mathcal{M}_g\right] = R\left[\mathcal{M}_T\right],$$
- leads to solution for torsion vector b^i as function of metric ϕ and arbitrary scalar field χ:
$$b^i = \varepsilon_{ik}\, \partial_k \phi + \partial_i \chi$$
- for $\chi = 0$: **autoparallels** of both manifolds equivalent up to reparameterization
- χ corresponds to local **roatations** of the frame field

Conclusion

- Autoparallels and curvature can be mapped
- Different distance measure
- Other tangent spaces and tensors

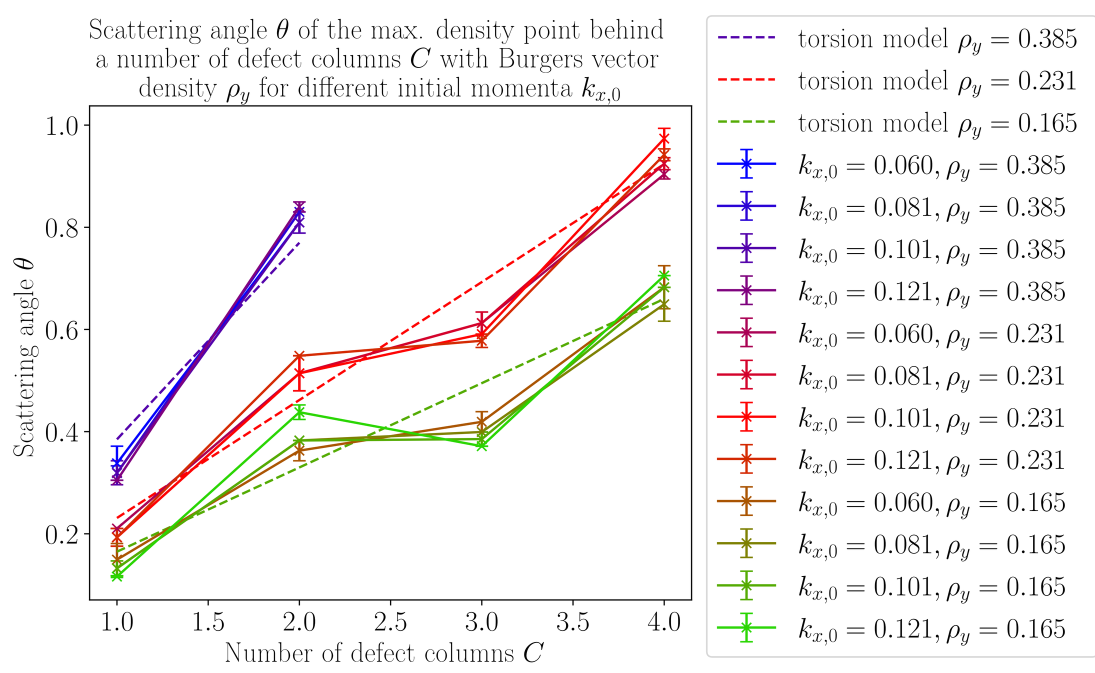
Tight-binding numerics

- **vector representation** of wavefunction at discrete lattice sites
- Hamiltonian is (sparse) **Matrix** acting on this vector to generate time evolution:
$$i\partial_t \vec{\psi} = H\vec{\psi} \quad \Rightarrow \quad \vec{\psi}(t) = e^{-iHt}\vec{\psi}_0$$



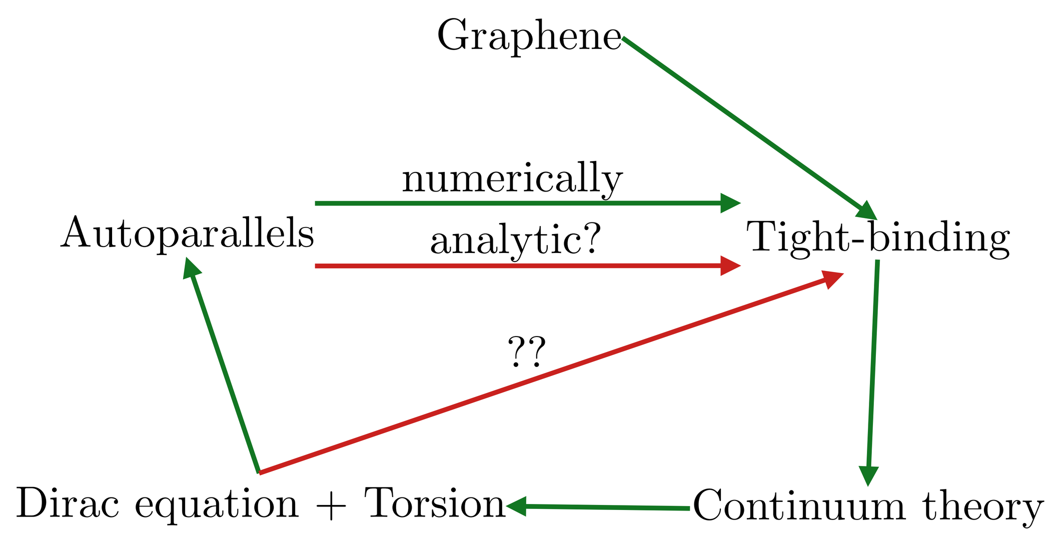
Torsion model

- $\mathcal{M}_T$ with $b^1(x)$ and $b^2 \equiv 0$
- Autoparallel equation:
$$\ddot{x} = -\dot{x}\dot{y}b^1, \quad \ddot{y} = \dot{x}^2 b^1$$
- Allows calculating the **scattering angle**:
$$\theta = \int b^1(x)\, dx$$
- Interpreting torsion as Burgers vector density:
$$\rho_y C = \frac{C \cdot (\vec{b})_x}{\Delta y} \approx \frac{\iint b^1 dx dy}{\Delta y} = \int b^1 dx = \theta$$



- Torsion model predicts linear relation $\theta = \rho_y C$
- Tight-binding simulations match prediction from torsion model with autoparallels

Conclusions & Outlook[4]



- Can the torsion interpretation of defects via autoparallels be shown analytically?
- Is it possible to describe the tight-binding results with the Dirac equation coupled to torsion?
- Does the physical mapping between torsion and metric apply to the description via the Dirac equation?

References

[1] Stegmann T, Szpak N. Current flow paths in deformed graphene. *New J. Phys.* **18**, 053016 (2016).
[2] Bhatt MD, Kim H, Kim G. Various defects in graphene: a review. *RSC Adv.* **12**, 21520 (2022).
[3] Heisenberg L. A systematic approach to generalisations of General Relativity. *Phys. Rep.* **796**, 1 (2019).
[4] de Juan F, Cortijo A, Vozmediano MAH. Dislocations and torsion in graphene. *Nucl. Phys. B* **828**, 625 (2010).